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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

Visitor Entry Logging and Security Monitoring System

Submitted by

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Semester: **II**

DECLARATION

I hereby declare that the project entitled **"VISITOR ENTRY LOGGING AND SECURITY MONITORING SYSTEM"** submitted in partial fulfillment of the requirements for the course **Java programming language** is a **record of my original work** carried out under the guidance of **Dr.Boopalan Sundaramoorthy**, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

Nivetha.K

CERTIFICATE

This is to certify that the project entitled **"Visitor Entry Logging and Security Monitoring System"** is a **bona fide work** carried out by **Nivetha.K, I B.Sc CS, Department Of Computer Science**, **KG College of Arts & Science**, in partial fulfillment of the requirements for the course **Java Programming Language**.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMABATORE

Date:10/04/2026

Submitted for the Viva-Voce Examination held on **10/04/2026**

Guide:
(Technical Trainer, KGCAS)

Head of the Department
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ABSTRACT

The **Visitor Entry Logging and Security Monitoring System** is a Java-based application developed to improve security and efficiently manage visitor records in organizations, offices, and institutions. In many places, visitor details are still recorded manually in logbooks, which can lead to data loss, inaccurate records, and difficulty in monitoring visitor movements. This project aims to provide a digital solution that automates the visitor registration and monitoring process.

The system is developed using **Java Swing** for the graphical user interface and integrates technologies such as **QR Code generation, camera-based photo capture, PDF report generation, and database-style table management**. When a visitor arrives, their details such as name, phone number, and purpose of visit are entered into the system. The application captures the visitor's photo using the system camera and generates a unique QR code for identification.

The QR code is used to record both **entry and exit times** of visitors by scanning it during arrival and departure. The system stores all visitor information in a table, allowing administrators to easily view, filter, and manage records. Additional features such as **dark mode interface, date-based filtering, and PDF report export** help administrators analyze and maintain visitor logs efficiently.

The main objective of this system is to enhance security, maintain accurate visitor records, and reduce manual effort in managing visitor entries. By automating the visitor management process, the system improves monitoring, ensures better data organization, and provides a reliable record for security purposes.

This project demonstrates the practical implementation of **Java programming, graphical user interface design, QR code technology, camera integration using OpenCV, and PDF report generation using iText**, making it a useful solution for modern visitor management systems.

1. INTRODUCTION

In many organizations, offices, and institutions, maintaining visitor records is an important part of ensuring security and proper monitoring of people entering and leaving the premises. Traditionally, visitor details are recorded manually in logbooks at the security desk. This manual method requires visitors to write their personal details such as name, phone number, purpose of visit, and entry time. However, this process is time-consuming and often leads to issues such as incomplete information, difficulty in searching past records, and errors in maintaining accurate visitor logs.

With the advancement of computer technology, digital visitor management systems have become an effective solution for improving security and record management. A computerized visitor entry system allows security staff to quickly register visitor details, capture visitor photos, and store the information in an organized digital format. This makes it easier to track visitor movements and maintain accurate records for future reference.

The Visitor Entry Logging and Security Monitoring System is a Java-based application designed to automate the process of visitor registration and monitoring. The system allows administrators or security staff to enter visitor details such as name, phone number, and purpose of visit. The application also captures the visitor's photo using the system camera and generates a unique QR code for each visitor. This QR code can be used to record both entry and exit times, making the tracking process faster and more efficient.

The system stores visitor information in a structured table format, enabling administrators to easily view, filter, and manage visitor records. Additional features such as QR code scanning, date-based filtering, dark mode interface, and PDF report generation help improve usability and data management.

This project is developed using Core Java and Java Swing for the graphical user interface. It also integrates technologies such as QR code generation using ZXing, camera integration using OpenCV, and PDF report generation using iText. The project demonstrates the practical implementation of Java programming concepts including classes, objects, event handling, file handling, and GUI development.

The main objective of this system is to improve security monitoring, reduce manual record-keeping, and provide an efficient and reliable way to manage visitor entries in an organization. By automating the visitor management process, the system helps ensure better record accuracy, faster visitor registration, and improved overall security management.

1.1 Objective

The main objective of the **Visitor Entry Logging and Security Monitoring System** is to develop a simple and efficient application using Core Java that automates the process of visitor registration, monitoring, and record management in organizations or institutions.

The specific objectives of this project are:

- To design and develop a **Java-based visitor management system** for recording visitor details digitally.
- To automate the **visitor entry and exit logging process** using QR code technology.
- To capture and store **visitor photographs using the system camera** for security purposes.
- To reduce manual errors and improve the **accuracy of visitor record management**.
- To provide a **user-friendly interface** for administrators or security staff to easily manage visitor information.
- To enable **QR code generation for visitor identification and quick scanning** during entry and exit.
- To generate and export **visitor reports in PDF format** for record keeping and analysis.
- To demonstrate the implementation of **Core Java programming concepts** such as classes, objects, event handling, file handling, GUI development using Java Swing, and integration with external libraries like OpenCV, ZXing, and iText.

The overall goal of this system is to **enhance security, improve visitor monitoring, and provide an efficient digital solution for managing visitor records**.

1.2 Overview

The Visitor Entry Logging and Security Monitoring System is a user-friendly application developed using the Java programming language. The system provides a digital platform for managing and monitoring visitor entries in organizations, offices, or institutions. It replaces the traditional manual visitor logbook with a computerized system that records visitor details efficiently and securely.

The system allows security staff or administrators to enter visitor information such as name, phone number, and purpose of visit. Once the details are entered, the application can capture the visitor's photograph using the system camera and generate a unique QR code for identification. This QR code can be used to record the visitor's entry and exit times by scanning it during arrival and departure.

All visitor information is stored and displayed in a table format within the system, allowing administrators to easily view and manage records. The system also provides features such as date-based filtering to view visitor logs for a specific time period. Additionally, the application allows users to export visitor records into a PDF report for documentation and security purposes.

The system simplifies visitor management by eliminating manual record keeping and reducing the chances of missing or incorrect information. The application is developed using Java Swing for the graphical user interface, along with external libraries for QR code generation, camera integration, and PDF report creation.

This project demonstrates the practical application of Java programming in developing real-world security management systems that improve efficiency, accuracy, and overall visitor monitoring.

1.3 Features

The **Visitor Entry Logging and Security Monitoring System** includes the following main features:

- **Admin Login System** – Provides secure access to the system through an administrator login to ensure that only authorized users can manage visitor records.
- **Visitor Registration** – Allows the administrator or security staff to enter visitor details such as name, phone number, and purpose of visit.
- **Camera Photo Capture** – Captures the visitor's photograph using the system camera to maintain a visual record for security purposes.
- **QR Code Generation** – Generates a unique QR code for each visitor that can be used for identification and tracking.
- **Entry and Exit Logging** – Records the visitor's entry and exit time using QR code scanning, helping to monitor the duration of each visit.
- **Visitor Records Table** – Displays all visitor information in a structured table format for easy viewing and management.
- **Date-Based Filtering** – Allows administrators to filter visitor records based on options such as today or this week.
- **PDF Report Generation** – Enables exporting visitor details and photos into a PDF report for documentation and record keeping.
- **Dark Mode Interface** – Provides a dark mode option for better visual comfort and improved user experience.
- **User-Friendly Interface** – Offers a simple and easy-to-use graphical interface developed using Java Swing for efficient visitor management

2.PROGRAM ANALYSIS

Program analysis is an important phase in software development where the problem is studied in detail and the system requirements are identified. This phase helps in understanding how the proposed system should function and what features are needed to solve the existing problems effectively.

The analysis stage focuses on identifying the limitations of the current system and designing a computerized solution that improves efficiency, security, and data management. By analyzing the existing process, developers can design a system that simplifies operations and reduces manual effort.

2.1 Problem Definition

In many organizations, offices, and institutions, visitor information is still recorded manually in logbooks at the security desk. Visitors are required to write their personal details such as name, phone number, purpose of visit, and entry time in a register. While this method has been used for many years, it creates several problems in managing visitor records.

Some common problems include:

- **Incomplete or inaccurate visitor information**
- **Difficulty in searching past visitor records**
- **Time-consuming manual entry process**
- **Lack of proper monitoring of visitor entry and exit times**
- **Risk of losing or damaging physical logbooks**

Manual visitor logging systems also make it difficult for security staff to track visitor movements and maintain proper records for future reference. In busy environments, this process can create delays and reduce the efficiency of security management.

Therefore, there is a need for a **digital visitor management system** that can record visitor details quickly and maintain accurate records. An automated system can also improve security by capturing visitor photos and tracking entry and exit times.

The **Visitor Entry Logging and Security Monitoring System** is designed to address these issues by providing a computerized solution that automates visitor registration, generates QR codes for visitor identification, records entry and exit times, and maintains organized digital records for easy access and reporting.

2.2 Project Overview

The **Visitor Entry Logging and Security Monitoring System** is a Java-based application developed to improve the process of managing and monitoring visitor records in organizations, offices, and institutions. The system replaces the traditional manual logbook method with a computerized solution that records visitor information quickly and efficiently.

The application provides a graphical user interface where administrators or security staff can enter visitor details such as name, phone number, and purpose of visit. The system also captures the visitor's photograph using the system camera and generates a unique QR code for identification. This QR code is used to record the visitor's entry and exit times when scanned during arrival and departure.

All visitor records are stored and displayed in a table within the system, making it easy for administrators to view and manage visitor data. The system also allows filtering of visitor records based on specific time periods such as today or this week. Additionally, the system can generate a PDF report containing visitor details and photos for documentation purposes.

The system includes the following functionalities:

- Entering and storing visitor details
- Capturing visitor photographs using the system camera
- Generating QR codes for visitor identification
- Recording visitor entry and exit times
- Displaying visitor records in a table format
- Filtering visitor data based on date
- Exporting visitor records to a PDF report

This project is implemented using **Core Java and Java Swing for the graphical user interface**, along with external libraries for **QR code**

generation, camera integration, and PDF report generation. The system

demonstrates how Java programming can be used to develop practical applications for security monitoring and visitor management

2.3 Scopes and Limitations

Scope

The scope of the **Visitor Entry Logging and Security Monitoring System** includes:

• Recording visitor details such as name, phone number, and purpose of visit
• Capturing visitor photos using the system camera
• Generating QR codes for visitor identification
• Recording visitor entry and exit times
• Displaying visitor records in a table
• Exporting visitor data as a PDF

report The system can be used in:

• Offices and companies
• Schools and colleges
• Hospitals
• Government buildings
• Apartments and institutions

Limitations

• Works only on a **single computer system**
• Does not use a **permanent database** for storing records
• QR scanning is **simulated** in the system

3.SYSTEM DESIGN

System design is the stage where the structure and components of the system are planned before implementation. In this phase, the system is divided into different modules and the workflow of the application is defined. Proper system design helps organize the program efficiently and makes development easier.

The **Visitor Entry Logging and Security Monitoring System** is designed as a Java-based application that allows administrators to register visitors, capture photos, generate QR codes, and monitor visitor entry and exit details.

The system is divided into several modules, where each module performs a specific function.

3.1 Modules

The main modules of the **Visitor Entry Logging and Security Monitoring System** are explained below:

1. Admin Login Module

This module provides secure access to the system. Only authorized administrators can log in using a username and password before accessing the visitor management features.

Functions of this module include:
- Verifying username and password
- Allowing access only to authorized users

2. Visitor Registration Module

This module allows the administrator to enter visitor details into the system.
Functions of this module include:
- Entering visitor name
- Entering phone number
- Entering purpose of visit
- Storing visitor information temporarily in the system

3. Photo Capture Module
This module captures the visitor's photo using the system camera.
Functions of this module include:

- Opening the system camera
- Capturing the visitor image
- Displaying the photo in the application

The captured photo is used for security verification and record keeping.

4. QR Code Generation Module
This module generates a unique QR code for each visitor based on their details.
Functions of this module include:

- Creating a QR code using visitor information
- Displaying the generated QR code in the system

The QR code is used for tracking visitor entry and exit.

5. Entry and Exit Logging Module
This module records the visitor's entry and exit times.
Functions of this module include:

- Recording entry time when the visitor arrives
- Recording exit time when the visitor leaves
- Updating visitor records in the table

6. Report Generation Module
This module generates visitor reports.
Functions of this module include:

- Displaying visitor records in a table
- Exporting visitor data and photos to a PDF report
- Saving the report for future reference

7. Exit Module

This module allows the administrator to exit the application safely.

When the exit option is selected, the system closes the program and stops all running processes.

4. METHODOLOGY

Methodology describes the process followed to develop the project. It explains the steps taken from understanding the problem to designing, implementing, testing, and documenting the system.

For the **Visitor Entry Logging and Security Monitoring System**, a structured development approach was followed to ensure the system works efficiently and meets security requirements.

The development process includes the following stages:

- Problem Analysis
- System Design
- Program Development
- Testing and Validation
- Result and Documentation

4.1 Problem Analysis

In this phase, the problem of manual visitor management was studied and analyzed. Many organizations still maintain visitor details using logbooks, which creates several issues.

The main problems identified are:

- Manual entry of visitor details takes more time
- Difficulty in tracking visitor entry and exit times
- Visitor records may be incomplete or difficult to search
- Risk of losing or damaging physical logbooks

Based on these problems, the system requirements were defined to develop a digital visitor management system that records and manages visitor details efficiently.

4.2 System Design

In this stage, the overall structure of the system was designed before coding began.

The system is divided into modules such as:

- Admin Login Module
- Visitor Registration Module
- Photo Capture Module
- QR Code Generation Module
- Entry and Exit Logging Module
- Report Generation Module

The program flow was designed so that administrators can easily register visitors, capture photos, generate QR codes, and manage visitor records.

4.3 Program Development

After designing the system, the application was developed using the Java programming language.

The following steps were followed during development:

- Creating classes and methods for the system
- Designing the graphical user interface using Java Swing
- Implementing QR code generation for visitor identification

• Integrating camera functionality to capture visitor photos
• Developing features for entry and exit logging
• Generating visitor reports in PDF format

Core Java concepts such as classes, objects, event handling, loops, conditional statements, and file handling were used during the development of the system.

4.4. Testing and Validation

Testing ensures that the system works correctly and performs all functions as expected. Different test cases were used to verify the features of the Visitor Entry Logging and Security Monitoring System.

Example Test Cases:

Test Case	Expected Result
Enter visitor details	Details stored correctly
Capture visitor photo	Photo captured and displayed
Generate QR code	QR code generated successfully
Scan QR code	Entry or exit time recorded

This phase ensures that the system runs smoothly, records visitor information correctly, and performs all operations without errors.

4.5 Result and Documentation

After successful testing, the results of the system are documented. The system successfully records visitor details, captures photos, generates QR codes, and logs entry and exit times.

The results are demonstrated through program execution and screenshots showing the working of the system interface and visitor records.

Proper documentation is prepared to explain the **system design, implementation, modules, and functionality** of the Visitor Entry Logging and Security Monitoring System.

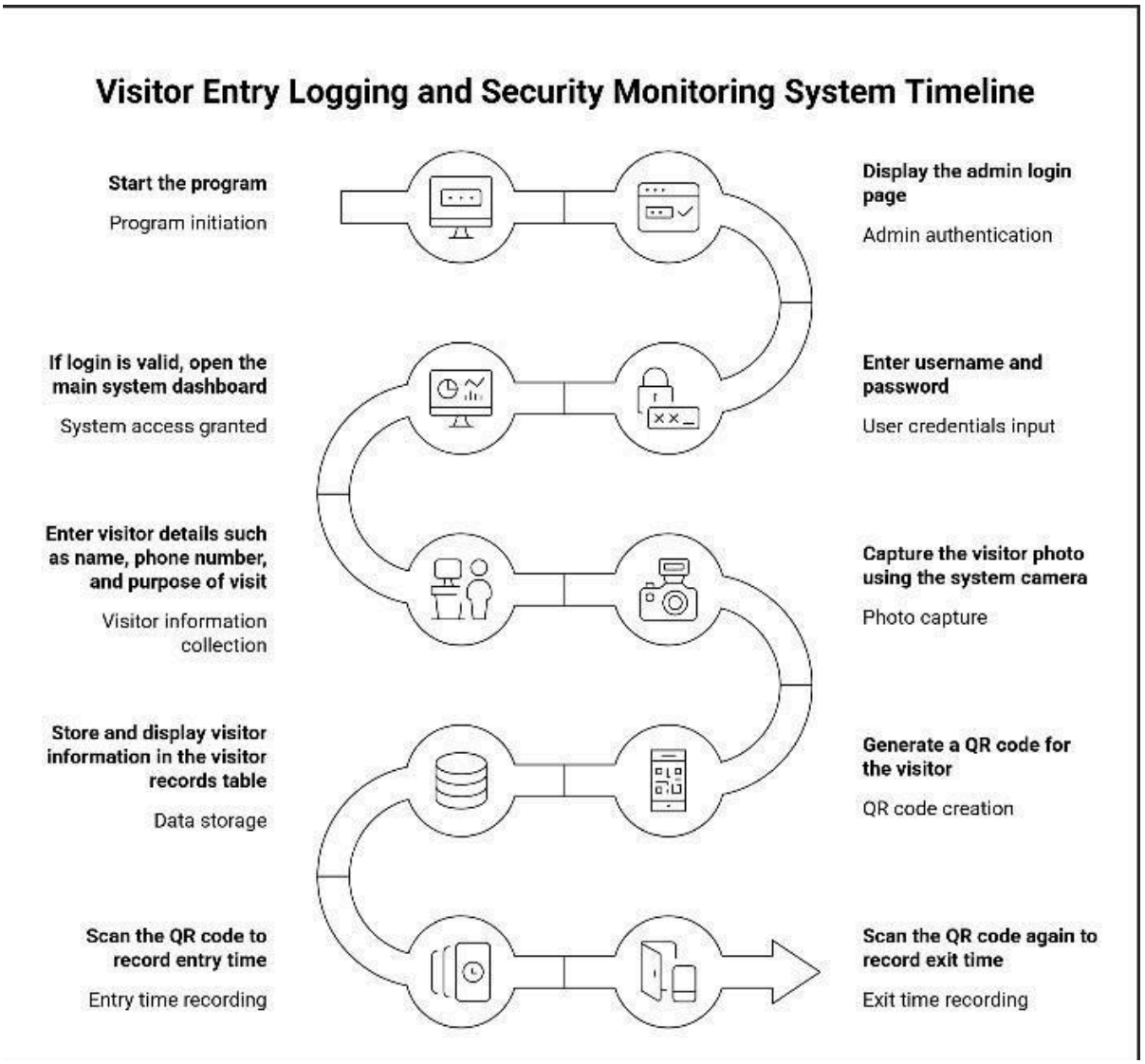
4.6. Algorithm

Algorithm for the **Visitor Entry Logging and Security Monitoring System:**

1. Start the program.
2. Display the admin login page.
3. Enter username and password.
4. If login is valid, open the main system dashboard.
5. Enter visitor details such as name, phone number, and purpose of visit.
6. Capture the visitor photo using the system camera.
7. Generate a QR code for the visitor.
8. Store and display visitor information in the visitor records table.
9. Scan the QR code to record entry time.
10. Scan the QR code again to record exit time.
11. Export visitor records as a PDF report if required.
12. Clear fields or continue registering new visitors.
13. End the program.

4.7. Flowchart

Students should draw a **flowchart showing the working process of the Visitor Entry Logging and Security Monitoring System**.



5 .SOURCE CODE

Main Class

```
public class VisitorApp extends JFrame {


  public static void main(String[] args) {

      new VisitorApp();

  }

}
```

Explanation:
This is the starting point of the program. When the program runs, it starts the VisitorApp and opens the application window.

Login Validation

```
if(user.getText().equals("admin") &&

 new String(pass.getPassword()).equals("1234")){


  login.dispose();

  initUI();

}
```

Explanation:
This code checks the username and password. If they are correct, the system opens the main visitor management screen.

QR Code Generation

```
QRCodeWriter writer = new QRCodeWriter();

BitMatrix matrix = writer.encode(lastQRData,

      BarcodeFormat.QR_CODE,150,150);
```

Explanation:
This code creates a QR code using the visitor information. The QR code is used to identify the visitor.

Entry Time Recording

```
model.addRow(new

  Object[]{ name, phone,

  purpose, currentTime(), ""

});
```

Explanation:
This code adds the visitor details and entry time to the table when the visitor enters.

Exit Time Recording

```
model.setValueAt(currentTime(), rowIndex, 4);
```

Explanation:
This code records the exit time when the visitor leaves the building.

6. INPUT AND OUTPUT SCREENS

Figure 1:Login

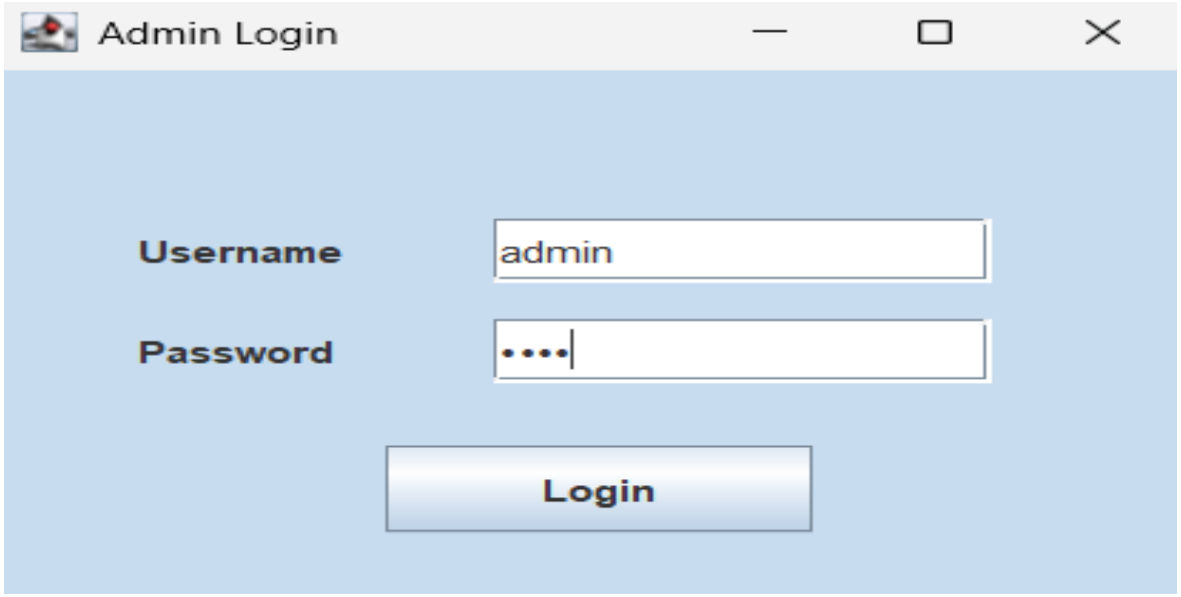


Figure 2:Main Menu Screens

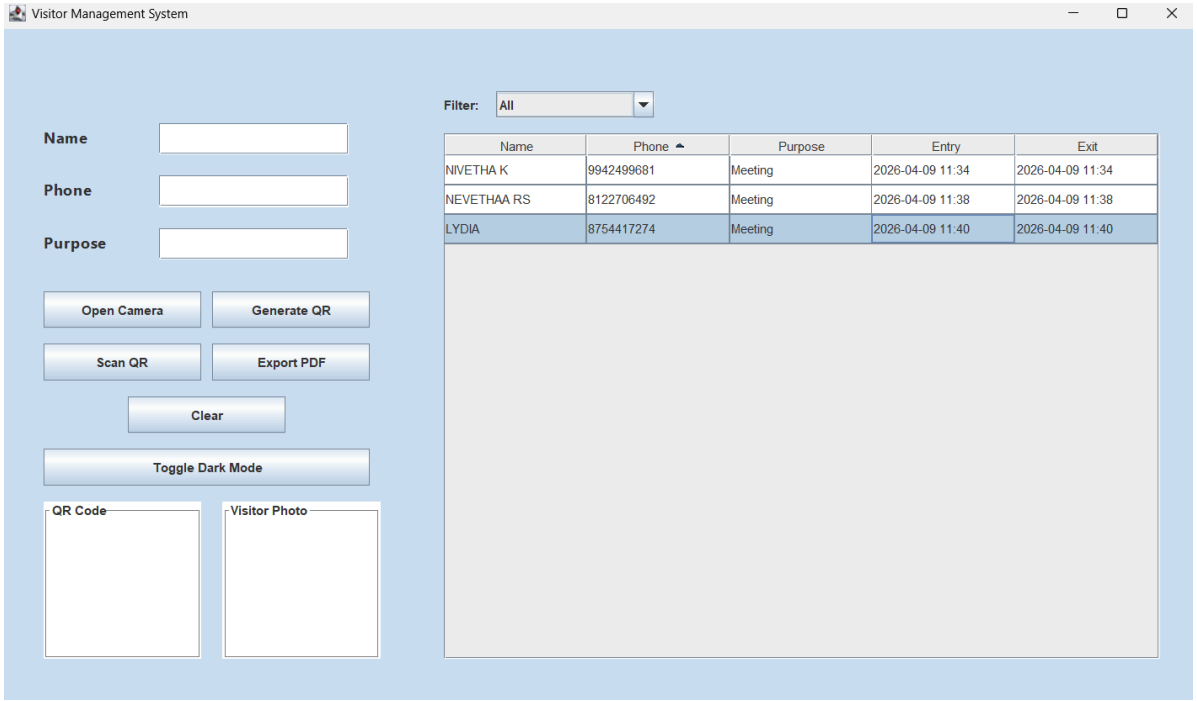


Figure 3:Visitor Log Viewing and Filtering Panel

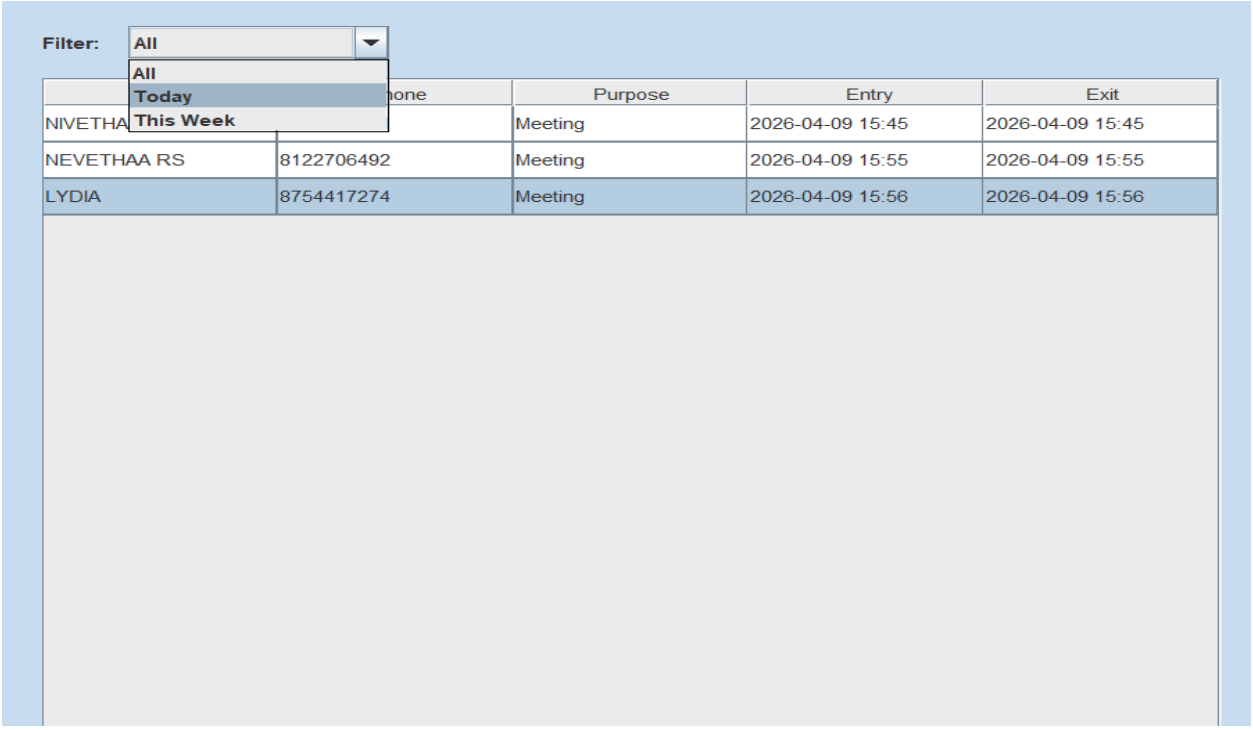


Figure 4:PDF is Saved

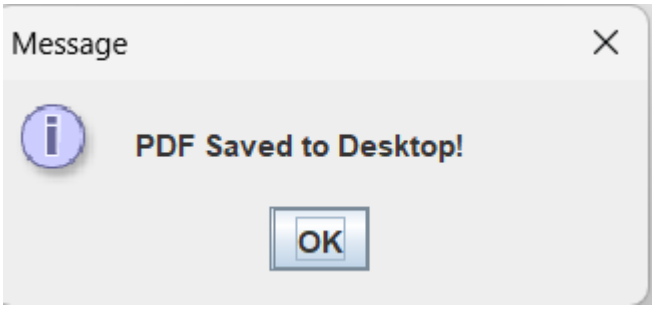
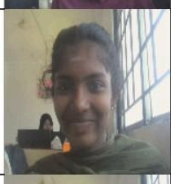
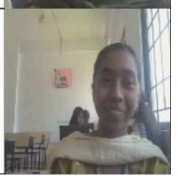


Figure 5:Visitors Records where stored in PDF

VISITOR REPORT

Name	Phone	Purpose	Entry	Exit	Photo
NIVETHA K	9942499681	Meeting	2026-04-09 11:34	2026-04-09 11:34	
NEVETHAA RS	8122706492	Meeting	2026-04-09 11:38	2026-04-09 11:38	
LYDIA	8754417274	Meeting	2026-04-09 11:40	2026-04-09 11:40	

7. DISCUSSION & CONCLUSION

7.1 Discussion

The **Visitor Entry Logging and Security Monitoring System** was developed to improve the process of managing visitor records in organizations. The system allows administrators to enter visitor details, capture visitor photos, generate QR codes, and record entry and exit times.

During the development of the project, Core Java concepts such as classes, methods, loops, conditional statements, and event handling were used. The system successfully stores visitor information and displays it in a table for easy monitoring.

The system was tested with different inputs to ensure that visitor details, QR code generation, and entry–exit logging work correctly. The results show that the system helps reduce manual work and improves visitor record management.

7.2 Conclusion

The **Visitor Entry Logging and Security Monitoring System** successfully demonstrates the practical use of Java programming in developing a real-world security application.

The system simplifies visitor management by recording visitor details, capturing photos, generating QR codes, and maintaining entry and exit records. It improves security and helps organizations maintain proper visitor logs.

This project helps students understand how Java programming can be used to develop practical software systems. The system can be further improved by adding features such as database storage, real-time QR scanning, and network-based access.

8. REFERENCES

The following resources were used during the development of this project:

Books

- Herbert Schildt – *Java: The Complete Reference*
- Kathy Sierra & Bert Bates – *Head First Java*
- Cay S. Horstmann – *Core Java Volume I: Fundamentals*

Websites

- https://www.oracle.com/java
- https://www.geeksforgeeks.org
- https://www.w3schools.com
- https://www.javatpoint.com